Problem Set 3 - IC104

Solutions

Essential Mathematics for Economic Analysis, 4th edition
Page 202. Question 2 a) b)

Find the derivative of:

(a) $y = x^3(\ln x)^2$; (b) $y = \dfrac{x^2}{\ln(x)}$

Answers:

$$a) x^2 \ln x(3 \ln x + 2)$$
$$b) \frac{x(2 \ln x - 1)}{(\ln x)^2}$$

Page 280. Question 4:

Find the production level Q, where Q is the number of units of product manufactured, that maximises profits when the Revenue function R(Q), and the Cost function C(Q) are as follows.

Hint: differentiate R(Q) to find the marginal profit per unit $R'(Q)$, differentiate C(Q) to find the marginal cost per unit $C'(Q)$, and then set $R'(Q) = C'(Q)$

(i) $R(Q) = 1840Q$; $C(Q) = 2Q^2 + 40Q + 5000$

(ii) $R(Q) = 2240Q$; $C(Q) = 2Q^2 + 40Q + 5000$

(iii) $R(Q) = 1840Q$; $C(Q) = 2Q^2 + 1940Q + 5000$

Answers:

$$Profit(Q) = R(Q) - C(Q), Profit'(Q) = R'(Q) - C'(Q) = 0$$

$a)$ $1840 - 4Q - 40 = 0 \Rightarrow Q^* = 450$

$b)$ $2240 - 4Q - 40 = 0 \Rightarrow Q^* = 550$

$c)$ $1840 - 4Q - 1940 = 0 \Rightarrow Q^* = -25$ Because negative production is impossible, $Q^* = 0$

Page 280. Question 6:

Let $R(Q) = PQ$ and $C(Q) = aQ^b + c$

where P, a, b, c are positive constants, and $b > 1$. Find the value of Q that maximises the profit:

$\pi(Q) = PQ - (aQ^b + c)$

(Use second order derivative to show why this is a maximum point)

Answers: $\pi(Q)$ is stationary at $Q = (\frac{P}{ab})^{\frac{1}{(b-1)}}$. Moreover, $\pi''(Q) = -ab(b-1)Q^{b-2} < 0$ for all $Q > 0$, so this is a maximum point.